

RAPITA HEMlib and MACH178 Prototype

RAPITA | ISOLDE | T4.3 / D4.5 Section 3.2.2 | Version ISOLDE-D4.5-2026.07

Field	Detail	Value
Schema Version		
		0.9
Basic Information		
IP Name		RAPITA HEMlib and MACH178 Prototype
Version		ISOLDE-D4.5-2026.07
Status		Demonstrated
Comment		IP Card for MACH178 prototype developmen and observability libraries (HEMlibs) tool implementation derived from ISOLDE D4.5.
Provider	Item 1 / Organization Name	Rapita Systems S.L.
Provider	Item 1 / Organization URL	https://www.rapitasystems.com/
Provider	Item 1 / Organization Alias	- RAPITA
License		Proprietary
Repository URL		https://www.rapitasystems.com/
Project		ISOLDE
Work Item		T4.3 / D4.5 Section 3.2.2
IP Type	Category	- Software Tool - Debugging, Performance, Visualization and Profiling Tools - Timing Analysis Software - V&V Tools
IP Type	Domain	- Safety
IP Type	Comment	Multicore timing observability and certification-oriented evidence generation
IP Type	Service	no
System-Level Information		
Architecture		RISC-V
Processor		NOEL-V
ISA Extension		Performance-monitoring counters and platform-specific HEM facilities
Board		Xilinx ZCU102
SoC		SELENE/NOEL-V platform with XNG
SW Dependencies		XNG hypervisor, compatible NOEL-V/SELENE bitstream, C toolchain and platform documentation
Requires Specialized Tool		Yes. Target platform integration and Rapita process/tool assets.
Interfaces and Artifacts		
CLI		- Compile HEMlib and RapiDaemons into the target image - Execute interference scenarios and collect HEM evidence

Field	Detail	Value
API		- HEMlib C functions for setup, initialization, reset and counter access
Input Artifacts	Item 1 / Artifact	Platform configuration and bitstream
Input Artifacts	Item 1 / Format	FPGA bitstream/configuration
Input Artifacts	Item 2 / Artifact	XNG integration
Input Artifacts	Item 2 / Format	Hypervisor package
Input Artifacts	Item 3 / Artifact	RapiDaemon stress workloads
Input Artifacts	Item 3 / Format	C source/binaries
Input Artifacts	Item 4 / Artifact	MACH178 procedures and checklists
Input Artifacts	Item 4 / Format	Documents/templates
Output Artifacts	Item 1 / Artifact	Hardware-event-monitor measurements
Output Artifacts	Item 1 / Format	Counter data
Output Artifacts	Item 2 / Artifact	Interference-channel characterization evidence
Output Artifacts	Item 2 / Format	Reports
Output Artifacts	Item 3 / Artifact	Certification-oriented procedures and checklists
Output Artifacts	Item 3 / Format	Documents
Output Artifacts	Item 4 / Artifact	RapiDaemon validation results
Output Artifacts	Item 4 / Format	Test evidence
Deployment		
Bundle		Proprietary C library, benchmark suite and certification-oriented process/documentation package
Project-Specific Features		
Purpose		Support multicore interference analysis and evidence for AMC 20-193 and DO-178C-oriented activities.
Validated Capabilities		- Interference-channel identification - Critical-configuration identification - Hardware-event-monitor identification and access - RapiDaemon stress generation - Automotive demonstrator integration
Limitations		- Restricted platform documentation limited some NOEL-V analysis depth. - Bitstream availability and extension support delayed parts of integration and testing.
Public Repository Status		Proprietary
Source Basis	Deliverable	ISOLDE D4.5 - Software for Hardware Design and Validation Implementation
Source Basis	Section	3.2.2 Tooling to ensure compliance with multicore timing certification
Source Basis	SW IP Card Model	Schema v0.9 with basicInfo, systemLevelInfo, interfaces, deployment and features
Requirements Summary	Count	16

Field	Detail	Value
Requirements Summary	IDs	- RAPITA-01 - RAPITA-02 - RAPITA-03 - RAPITA-04 - RAPITA-05 - RAPITA-06 - RAPITA-07 - RAPITA-08 - RAPITA-01 - RAPITA-02 - RAPITA-03 - RAPITA-04 - RAPITA-05 - RAPITA-06 - RAPITA-07 - RAPITA-08
Metadata Review	Partner Review Required	false
Metadata Review	Items	- No pending additions from our side.